# Mission Mausam HPC Optimization Manual & Unified Model Tuning

Standard Operating Benchmarks, Compiler Flags (Intel oneAPI / GCC), and NUMA Affinity Configurations for Ministry of Earth Sciences Clusters

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## 1. Computing Architecture & Partition Specifications

The National Supercomputing Mission infrastructure at IMD and IITM Pune deploys heterogeneous high-performance nodes featuring dual Intel Xeon Platinum processors coupled with NVIDIA H100 GPU tensor accelerators and Mellanox HDR 200Gbps InfiniBand interconnects in a fat-tree topology.

| Subsystem | Hardware Specification | Peak TFLOPs | Operational Target |
|---|---|---|---|
| Pratyush Main Compute | 39,000+ CPU Cores (Cray XC40) | 4,006 TFLOPs | Operational Daily Ensemble NWP (00Z & 12Z) |
| Mihir Cluster (NCMRWF) | 23,000+ CPU Cores | 2,800 TFLOPs | Medium-Range Coupled Earth Systems |
| GPU AI/ML Partition | 128x NVIDIA H100 80GB SXM5 | 4,200 TFLOPs Tensor | Deep Learning Nowcasting & PINNs Inference |

## 2. Production Compiler Tuning & Vectorization

When compiling thermodynamic and microphysics parameterizations, AVX-512 vectorization must be combined with strict IEEE floating-point compliance to prevent numerical drift in long-range precipitation accumulations.

```
# Production Compiler Flags for Intel Fortran (ifort / ifx 2024.1)
FCFLAGS = -O3 -xCORE-AVX512 -qopt-zmm-usage=high -fp-model precise \
-qopenmp -align array64byte -heap-arrays 64 -mcmodel=medium
```

## 3. OpenMP Hybrid Thread Pinning & NUMA Architecture

Deploying hybrid MPI + OpenMP models requires binding OpenMP threads to physical processor cores on the same NUMA socket to prevent memory bus contention across sockets: export OMP_PLACES=cores; export OMP_PROC_BIND=close.